

1. Suppose X is a Binomial random variable with parameters $n = 5$ and $p = 1/3$. Suppose that Y has a uniform distribution on the integers from 0 to 5, inclusive. Also suppose that X and Y are independent.

1a. Find $P(X = Y)$.

1b. Find $P(X > Y)$.

2. Suppose that 5 people each bring a book to a book club. During a break at the book club, they stop reading and they eat lunch. The fire alarm rings during lunch, so they are forced to each grab a book at random, as they hurry for the door to escape. Suppose that all assignments of books to people are equally likely.

Let X denote the number of people who get their correct book, during the rush to escape.

2a. Find the expected value of X .

2b. Find the variance of X .

3. Consider a bag of 10 red bears, 10 blue bears, and 10 yellow bears. Suppose that 10 students each choose 3 bears, without replacement. Let X denote the number of students who get 3 bears of the same color in their selection.

3a. Find the expected value of X .

3b. Find the variance of X .

4. Suppose that 2 red beads, 3 blue beads, and 4 yellow beads are placed randomly onto a circular necklace, with all configurations equally likely. Find the probability that all of the beads of each color are grouped together with all of the other beads of that same color, i.e., the reds are together, the blues are together, and the yellows are together.

5. A bag of candy contains 10 green M&M's and 10 red M&M's. Suppose that Alice randomly eats M&M's (with all selections equally likely) until the bag contains only 1 color of remaining M&M's. What is the expected number of M&M's in the bag at that point?